Akka vs. n8n

A comparison for teams building agentic AI

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n8n automates your workflows; the Akka Agentic AI Platform runs your production agentic systems — and guarantees resilience and scalability. n8n is excellent visual workflow automation with AI Agent nodes — the fast path for integrations and internal automations. It is not built for production agentic scale, durable execution, or AI governance: the heavy-load, reliability, and compliance burden stays with you.

99.5%	99.9999%	4ms	5B
N8N ENTERPRISE SLA	AKKA SLA	AKKA STATE READS	TOKENS / SEC BENCHMARK

DIMENSION	N8N	AKKA
What it is	A visual workflow-automation platform with AI Agent nodes	The Akka Agentic AI Platform — guarantees resilience and scalability
Scope	Workflow automation and integrations; agentic scale, durable execution, and governance are the customer's burden	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.5% uptime (Enterprise Cloud), excluding planned maintenance	99.9999% — entire platform, backed by indemnities
HA / DR	Queue mode + multi-main, Redis required; no published RTO/RPO or active-active multi-region guarantee	Sub-1-minute RTO; zero-byte RPO; active-active across regions
Durable execution	Execution state persisted to PostgreSQL; in-flight runs on a crashed worker can be marked "crashed" — no deterministic replay substrate	Event-sourced; the runtime handles recovery and replay
AI agents	AI Agent node built on the LangChain JS framework	Native agents with tools, handoffs, guardrails, interaction logging
Memory	Bolt-on sub-nodes; external store you provision	Durable in-memory, 4ms reads / sub-10ms writes
Governance / EU AI Act	Audit logs and log streaming (Enterprise); no inline policy enforcement, classification, or sealed audit artifact	Aspect-woven runtime enforcement + full pre-production governance
Cost model	Per-execution metering (€24–€800/mo by plan; Enterprise custom) + infra you operate when self-hosted	Shared compute; up to 90% lower infrastructure for the same workload
Certifications	SOC 2 Type II, GDPR (trust.n8n.io)	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)
License	Sustainable Use License (fair-code, source-available, not open source); internal-business-purpose only; reselling n8n as a service prohibited; n8n Embed needs a commercial license	BSL; deploy anywhere including sovereign cloud; portable specs

n8n Is an Automation Tool; Akka Is an Agentic Platform

n8n is a visual workflow-automation platform that connects 400+ apps and APIs, with AI Agent nodes added on top. It is excellent for integrations and internal automations, and self-hosting gives teams full control of their data. A production agentic system needs more

than node-based automation — durable execution, horizontal agentic scale, durable memory, streaming, and runtime governance — and n8n leaves those to you.

Capability	n8n	Akka
Visual workflow automation + integrations	Yes (400+ integrations)	Programmatic; integrations via code
Native AI agents (tools, handoffs, streaming)	AI Agent node on LangChain JS	Built in
Durable execution / replay	PostgreSQL-persisted; no deterministic replay	Event-sourced, runtime-managed recovery
Durable memory	Bolt-on sub-nodes; external store	Built in, 4ms / sub-10ms
Real-time streaming	None native	Built in, backpressured, petabyte-scale
Inline governance / policy enforcement	None	Inline, runtime-embedded
Pre-production governance	None	Classification, sign-offs, sealed posture

n8n's AI Agent node is a wrapper over the LangChain JavaScript framework. It is ideal for triggering an LLM step inside an automation; it is not an agent runtime engineered for unattended, process-attached execution at scale.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform generates it, tests it, and runs it as one certified service.

Reliability, Scale, and Durable Execution

n8n Enterprise Cloud publishes a 99.5% uptime commitment (excluding planned maintenance during 10am–5pm German time) — roughly 1.8 days of allowed downtime per year. Akka's contractual SLA is 99.9999% — about 31 seconds per year — across the entire platform, backed by indemnities.

Metric	n8n	Akka
Availability SLA	99.5% (Enterprise Cloud)	99.9999%
Allowed downtime / year	~1.8 days	~31 seconds
RTO	Not published	Sub-1 minute
RPO	Not published	Zero byte
HA model	Queue mode + multi-main; Redis + PostgreSQL required	Active-active across regions
Durable execution	PostgreSQL-persisted; crashed-worker runs can be marked "crashed"	Event-sourced, replayable from the journal

To scale horizontally, n8n runs in **queue mode**: a Redis message broker (Bull queue) distributes executions to worker processes, PostgreSQL is mandatory, and multiple "main" instances provide availability. This is capable for automation throughput, but n8n provides no deterministic durable-execution substrate — when a worker dies mid-execution, in-flight runs can be marked "crashed" rather than transparently replayed, and recovery is the operator's responsibility. Akka's actor-based runtime runs 4KB actors at 200M actors/core across clusters up to 1M nodes; state is durable in-memory and replayable from the event journal; the runtime owns recovery, so application code does not have to.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing.

n8n Cloud bills **per execution** (€24/mo Starter / 2,500 executions, up to €800/mo Business / 40,000 executions; Enterprise custom), and that meter covers automation only. Memory, streaming, agent infrastructure, and the database sit outside it — provisioned and operated by you when self-hosted. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

Governance and the EU AI Act

n8n Enterprise provides audit logs and log streaming — useful operational telemetry. It provides no AI governance for production agents: no inline policy enforcement, no decision explainability, no human pause/override of a running agent, no immutable interaction ledger, no pre-deployment classification, and no sealed audit artifact. The compliance burden stays with the customer.

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

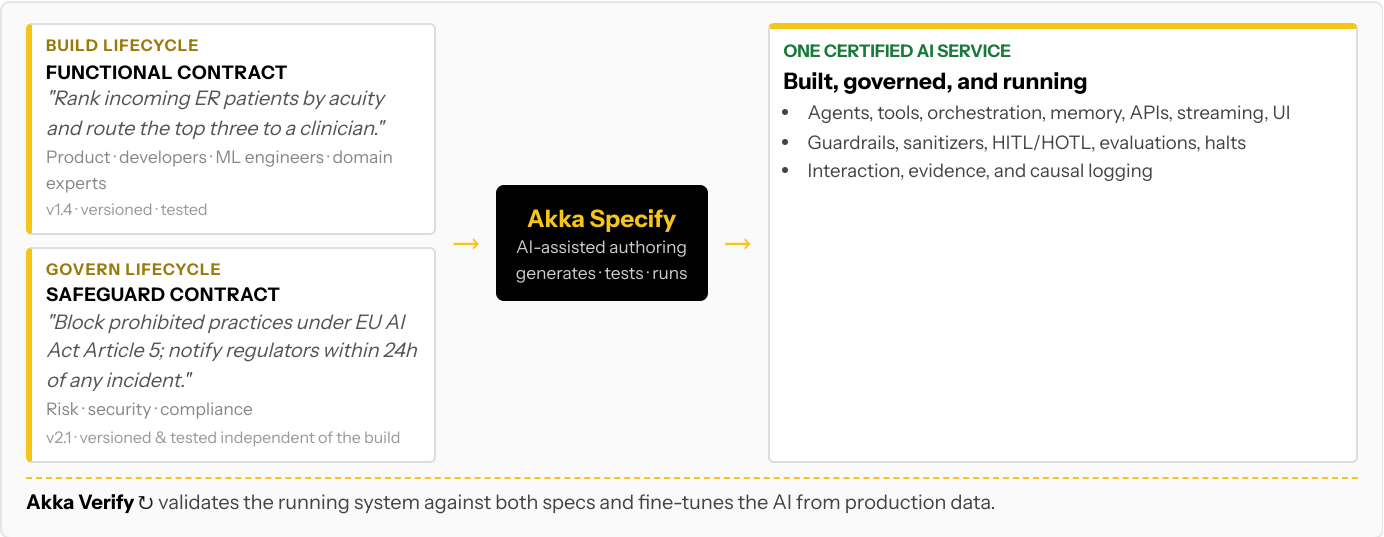
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka Governs Inline

The EU AI Act is enforceable now (prohibited practices since Feb 2025, high-risk since Aug 2025). Akka governs at the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; pre-deployment classification against **240 regulations and frameworks and 1,714 controls** (700 of which carry a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance n8n leaves to you, Akka enforces inline.

Two Lifecycles, One Certified System

Building on n8n means a workflow author wiring nodes on a canvas; there is no co-equal, independent governance lifecycle and no path for a risk officer to author and version enforceable safeguards. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow n8n has no equivalent for.

Real-Time Streaming at Petabyte Scale

n8n has no native streaming engine; real-time pipelines and agent feedback loops are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, Accountability, and Licensing

Buyer concern	n8n	Akka
Certifications & audits	SOC 2 Type II, GDPR (trust.n8n.io)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The automation layer; you operate scale, durability, and governance	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $180M Series C at a $2.5B valuation (Oct 2025); founded 2019; scaling on investor capital	Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Per-execution metering that scales with load	Fixed annual fee finance can forecast
License	Sustainable Use License — fair-code, source-available, not open source; internal-business-purpose only; reselling/hosting n8n as a service prohibited; embedding requires the n8n Embed commercial license	BSL; deploy on Akka cloud, hyperscaler VPC, your Kubernetes, on-prem, or sovereign cloud; portable specs

n8n is well-funded and growing fast — the decision is scope and accountability. n8n gives you a best-in-class automation canvas to integrate apps and trigger AI steps; Akka gives you the agentic platform, and owns the SLA, the durability, and the governance. The Sustainable Use License is also a procurement gate worth reading early: source-available but commercially restricted, with no right to offer n8n as a hosted service and a separate commercial license required to embed it.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already run n8n for our automations. Why add Akka?
Keep n8n for what it is excellent at — integrations and internal automations. When you move into production agentic AI you also need durable execution, horizontal agentic scale, durable memory, streaming, and runtime governance, which n8n does not provide. Akka runs the agentic systems on one platform while n8n keeps handling your automations.

n8n has AI Agent nodes and raised $180M at a $2.5B valuation. Doesn't that close the gap?
n8n's AI Agent node is a LangChain-based step inside a workflow, and strong funding reflects real demand for automation. But agent nodes on an automation engine are not a production agent runtime: there is no durable-execution substrate, no inline governance, and no platform-wide SLA. Capitalization does not add those layers.

Can we add governance on top of n8n?
You can add log streaming and audit logs, but the EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override on running processes, classification before a system ships, and a sealed audit artifact. Telemetry read after the fact cannot gate a deployment. Akka embeds all of this and covers pre-deployment governance.

Isn't n8n cheaper because it is free to self-host?
The n8n engine is source-available under the Sustainable Use License (fair-code, not open source), and self-hosting means you build and operate scale, HA/DR, durability, and everything around it. Cloud bills per execution. Akka's shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on a fixed annual fee —

and the license is a procurement gate: reselling n8n as a service is prohibited and embedding requires a commercial license.

Sources

n8n Enterprise Cloud SLA: n8n.io/legal/cloud-enterprise-terms — 99.5% uptime, excluding planned maintenance (10am–5pm German time)

n8n scaling / queue mode: docs.n8n.io/hosting/scaling/queue-mode — Redis (Bull queue) + worker processes, PostgreSQL required, multi-main for availability

n8n durable execution: community.n8n.io ("What happens if n8n worker crashes?") & github.com/n8n-io/n8n #22541 — crashed-worker executions can be marked "crashed"; recovery operator-managed

n8n AI Agent node: docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent — built on the LangChain JS framework

n8n pricing: n8n.io/pricing — per-execution metering, €24/mo Starter (2,500 exec) to €800/mo Business (40,000 exec); Enterprise custom

n8n license: docs.n8n.io/sustainable-use-license & blog.n8n.io/announcing-new-sustainable-use-license — fair-code, source-available, not open source; internal use only; no reselling as a service; n8n Embed commercial license

n8n security & certifications: n8n.io/legal/security & trust.n8n.io — SOC 2 Type II, GDPR

n8n funding: blog.n8n.io/series-c — $180M Series C at $2.5B valuation (Oct 2025); founded 2019

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 240 regulations and frameworks / 1,714 controls / 700 with a financial penalty; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital